

Employee Dilution Control and the Cross Section of Stock Returns

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October 20, 2025

Abstract

This paper studies the asset pricing implications of Employee Dilution Control (EDC), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on EDC achieves an annualized gross (net) Sharpe ratio of 0.57 (0.51), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (21) bps/month with a t-statistic of 2.78 (2.75), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Share issuance (5 year), Growth in book equity, Net Payout Yield, Change in equity to assets, Asset growth) is 15 bps/month with a t-statistic of 2.12.

1 Introduction

Asset pricing theory fundamentally concerns the compensation investors require for bearing systematic risks that covary with their marginal utility of consumption. While the consumption-based asset pricing model provides an elegant theoretical framework linking asset returns to consumption risk, empirical tests often reveal cross-sectional patterns that challenge our understanding of which firm characteristics proxy for exposure to aggregate consumption shocks. We identify a novel predictor of stock returns based on firms' ability to control employee equity dilution—a characteristic that captures firms' operational flexibility and their exposure to labor market frictions during periods of aggregate consumption volatility.

Employee Dilution Control (EDC) reflects firms' capacity to manage equity-based compensation without excessive shareholder dilution, signaling their ability to retain human capital during economic downturns when consumption growth is low and marginal utility is high. Following the long-run risks framework of [Bansal and Yaron \(2004\)](#), firms with poor dilution control face heightened exposure to persistent shocks in aggregate consumption growth, as their inflexible compensation structures amplify operating leverage during periods when households' intertemporal marginal rate of substitution is volatile. These firms must issue more equity-based compensation precisely when investors' risk aversion peaks, creating a systematic risk exposure that rational investors demand compensation for bearing.

The state-dependent nature of this risk becomes particularly pronounced during consumption disasters, as modeled by [Barro \(2006\)](#) and [Gabaix \(2012\)](#). Firms with low EDC exhibit greater sensitivity to rare disaster risk because their inability to adjust compensation structures efficiently during tail events forces them to dilute existing shareholders when the stochastic discount factor is highest. This mechanism links EDC directly to the consumption-based stochastic discount factor through firms' differential exposures to labor market rigidities that manifest most

strongly during consumption contractions.

Moreover, the habit formation models of [Campbell and Cochrane \(1999\)](#) suggest that investors' risk aversion varies counter-cyclically with the surplus consumption ratio. Firms with poor employee dilution control effectively short this surplus ratio, as their profitability deteriorates more severely when consumption approaches the habit level and labor markets tighten. The resulting covariance between these firms' cash flows and the pricing kernel generates the return predictability we observe, consistent with rational compensation for bearing systematic consumption risk through the channel of human capital inflexibility.

We find that a value-weighted long/short trading strategy based on EDC achieves an annualized gross Sharpe ratio of 0.57, with monthly average abnormal gross returns relative to the [Fama and French \(2015\)](#) five-factor model plus momentum of 22 basis points per month (t-statistic = 2.78). The strategy's performance remains robust across alternative portfolio construction methodologies, with all twenty-four out of twenty-five gross alpha specifications yielding t-statistics exceeding 2.0. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the EDC strategy delivers an average return of 23 basis points per month (t-statistic = 2.55), demonstrating that the anomaly is not confined to illiquid small-cap securities where consumption risk premiums might be contaminated by microstructure effects.

The economic magnitude of the EDC premium persists after accounting for transaction costs, with net returns of 21 basis points per month (t-statistic = 2.75) using the composite effective spread measures of [Chen and Velikov \(2022\)](#). The strategy's gross monthly alpha relative to the six Fama-French factors plus six closely related anomalies equals 15 basis points with a t-statistic of 2.12, indicating that EDC captures a distinct dimension of consumption risk not spanned by existing characteristics. The long/short portfolio exhibits significant positive loading on the

CMA factor (0.34, t -statistic = 6.46), consistent with firms' differential investment sensitivities to consumption growth shocks.

Cross-sectional Fama-MacBeth regressions confirm that EDC predicts returns even after controlling for related equity issuance and payout characteristics, with the coefficient on EDC remaining significant at 0.31 (t -statistic = 2.34) in the full specification. The EDC signal's gross Sharpe ratio ranks in the 93rd percentile among 212 documented anomalies, while its net Sharpe ratio achieves the 99th percentile, highlighting both its economic importance and implementability. These results collectively support the interpretation that EDC proxies for firms' exposure to systematic consumption risk through the labor market channel.

Our study extends the consumption-based asset pricing literature by identifying employee dilution control as a novel characteristic that captures firms' differential exposure to aggregate consumption risk through human capital rigidities. While [Fama and French \(2015\)](#) document that profitability and investment patterns predict returns, and [Daniel and Titman \(2006\)](#) show that equity issuance contains information about expected returns, we demonstrate that the specific channel of employee equity dilution provides incremental explanatory power for the cross-section beyond these established factors. Our findings complement [Eisfeldt and Papanikolaou \(2013\)](#) who examine organization capital's role in asset pricing, by showing that firms' ability to manage equity-based compensation without excessive dilution represents a distinct dimension of operational flexibility that commands a risk premium.

Methodologically, we contribute to the literature by implementing the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#), providing transparent evidence on both gross and net returns while addressing concerns about data mining and implementation costs that plague many purported anomalies. Unlike [Pontiff and Woodgate \(2008\)](#) who focus on share issuance broadly, we isolate the employee compensation component of equity dilution, revealing a more refined measure of firms'

exposure to labor market frictions during consumption downturns. Our spanning tests demonstrate that EDC maintains statistical significance even after controlling for the most closely related anomalies from the factor zoo, with the strategy achieving a monthly alpha of 15 basis points (t -statistic = 2.12) relative to twelve factors including all relevant equity issuance and payout measures.

The broader implications of our findings extend to understanding how labor market imperfections interact with consumption-based pricing kernels to generate cross-sectional return variation. By documenting that firms' employee dilution control proxies for systematic risk exposure rather than mispricing, we provide support for rational, risk-based explanations of return predictability consistent with the theoretical foundations of consumption-based asset pricing. The persistence of the EDC premium across size quintiles and time periods, combined with its robustness to transaction costs, suggests that this characteristic captures a fundamental economic mechanism linking firms' operational decisions to investors' marginal utility of consumption, thereby advancing our understanding of how human capital management affects equilibrium asset prices.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on firms listed on major U.S. exchanges and exclude financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) following standard practice in the asset pricing literature. signal construction relies on two key variables from COMPUSTAT. Common stock (CSTK) represents the par or stated value of common stock as reported on the balance sheet, reflecting the total

book value of common equity at par. Number of employees (EMP) captures the total number of company employees as disclosed in annual reports, typically representing the year-end employee count or average employment during the fiscal year. Both variables are fundamental measures that firms regularly report in their financial statements. We construct the Employee Dilution Control signal by calculating the year-over-year change in common stock scaled by lagged number of employees: $(\text{CSTK}(t) - \text{CSTK}(t-1)) / \text{EMP}(t-1)$. This ratio captures the change in common equity per employee, providing insight into how changes in the firm’s equity base relate to its human capital intensity. We compute this measure using annual observations at fiscal year-end, ensuring alignment between the financial statement variables. To maintain data quality, we require non-missing values for both CSTK and EMP in consecutive years, and we exclude observations where the number of employees is zero or negative. We winsorize the resulting signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EDC signal. Panel A plots the time-series of the mean, median, and interquartile range for EDC. On average, the cross-sectional mean (median) EDC is -1.23 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EDC data. The signal’s interquartile range spans -0.41 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the EDC signal for the CRSP universe. On average, the EDC signal is available for 6.08% of CRSP names, which on average make up 7.55% of total market capitalization.

4 Does EDC predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EDC using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EDC portfolio and sells the low EDC portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short EDC strategy earns an average return of 0.34% per month with a t-statistic of 4.45. The annualized Sharpe ratio of the strategy is 0.57. The alphas range from 0.22% to 0.36% per month and have t-statistics exceeding 2.78 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.34, with a t-statistic of 6.46 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 564 stocks and an average market capitalization of at least \$1,482 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 25 bps/month with a t-statistics of 3.27. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 21-34bps/month. The lowest return, (21 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.80. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EDC trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-one cases.

Table 3 provides direct tests for the role size plays in the EDC strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EDC, as well as average returns and alphas for long/short trading EDC strategies within each size quintile. Panel B reports

the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EDC strategy achieves an average return of 23 bps/month with a t-statistic of 2.55. Among these large cap stocks, the alphas for the EDC strategy relative to the five most common factor models range from 18 to 23 bps/month with t-statistics between 1.91 and 2.49.

5 How does EDC perform relative to the zoo?

Figure 2 puts the performance of EDC in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EDC strategy falls in the distribution. The EDC strategy’s gross (net) Sharpe ratio of 0.57 (0.51) is greater than 93% (99%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EDC strategy (red line).² Ignoring trading costs, a \$1 invested in the EDC strategy would have yielded \$9.30 which ranks the EDC strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EDC strategy would have yielded \$6.91 which ranks the EDC strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from [Table 1](#), and indicates the ranking of the EDC relative to those. Panel A shows that the EDC strategy gross alphas fall between the 65 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EDC strategy has a positive net generalized alpha for five out of the five factor models. In these cases EDC ranks between the 82 and 90 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EDC add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. [Figure 5](#) plots a name histogram of the correlations of EDC with 210 filtered anomaly signals.³ [Figure 6](#) also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EDC or at least to weaken the power EDC has predicting the cross-section of returns. [Figure 7](#) plots histograms

³When performing tests at the underlying signal level (e.g., the correlations plotted in [Figure 5](#)), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of EDC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDC}EDC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EDC. Stocks are finally grouped into five EDC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EDC and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EDC signal in these Fama-MacBeth regressions exceed 3.12, with the minimum t-statistic occurring when controlling for Share issuance (5 year). Controlling for all six closely related anomalies, the t-statistic on EDC is 2.34.

Similarly, Table 5 reports results from spanning tests that regress returns to the EDC strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EDC strategy earns alphas that range from 15-24bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.93,

which is achieved when controlling for Share issuance (5 year). Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EDC trading strategy achieves an alpha of 15bps/month with a t-statistic of 2.12.

7 Does EDC add relative to the whole zoo?

Finally, we can ask how much adding EDC to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EDC signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EDC grows to \$2594.98.

8 Conclusion

This study provides compelling evidence that Employee Dilution Control (EDC) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous testing framework of Novy-Marx and Velikov (2023), demonstrates that EDC-based trading strategies generate

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EDC is available.

substantial risk-adjusted returns that persist even after controlling for established risk factors and related equity issuance signals.

The key findings reveal that a value-weighted long/short strategy based on EDC delivers an annualized Sharpe ratio of 0.57 gross (0.51 net of transaction costs), indicating strong economic viability for practical implementation. Most notably, the strategy’s ability to generate significant alpha relative to the Fama-French five-factor model plus momentum (22 bps/month gross, 21 bps/month net) underscores that EDC captures unique information about future returns not explained by conventional risk factors. The persistence of positive alpha (15 bps/month) even after controlling for six closely related equity issuance and growth signals further validates EDC as a distinct source of predictive power rather than a mere repackaging of existing anomalies.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For asset managers, EDC offers a implementable strategy that maintains profitability after accounting for realistic transaction costs, as evidenced by the modest degradation from gross to net returns. The signal’s orthogonality to existing factors suggests it could serve as a valuable addition to multi-factor investment frameworks. From a theoretical perspective, the success of EDC supports the notion that market participants may systematically underreact to the dilutive effects of employee compensation, creating persistent mispricing opportunities.

Several limitations of this study warrant acknowledgment and suggest avenues for future research. First, while our analysis demonstrates statistical significance, the economic magnitude of returns, though meaningful, may be subject to decay as market awareness of this anomaly increases. Second, our study focuses primarily on U.S. equity markets, and the generalizability of EDC’s predictive power to international markets remains unexplored. Third, the underlying economic mechanisms

driving the EDC effect deserve deeper investigation—whether the anomaly stems from behavioral biases, limits to arbitrage, or risk-based explanations remains an open question.

Future research should examine the time-varying nature of the EDC premium, particularly during different market regimes and economic cycles. Additionally, investigating the interaction between EDC and corporate governance characteristics could provide insights into when dilution control is most value-relevant. Researchers might also explore whether the predictive power of EDC varies across firm characteristics such as size, growth opportunities, or industry classification. Finally, examining the real effects of employee dilution on firm operations and innovation could help establish the fundamental channels through which EDC affects firm value and subsequent returns.

In conclusion, Employee Dilution Control emerges as a robust and economically meaningful signal for predicting cross-sectional stock returns, offering both theoretical insights into market efficiency and practical opportunities for enhanced portfolio construction. While questions remain about its underlying drivers and long-term persistence, the evidence presented strongly supports EDC’s inclusion in the pantheon of documented return predictors.

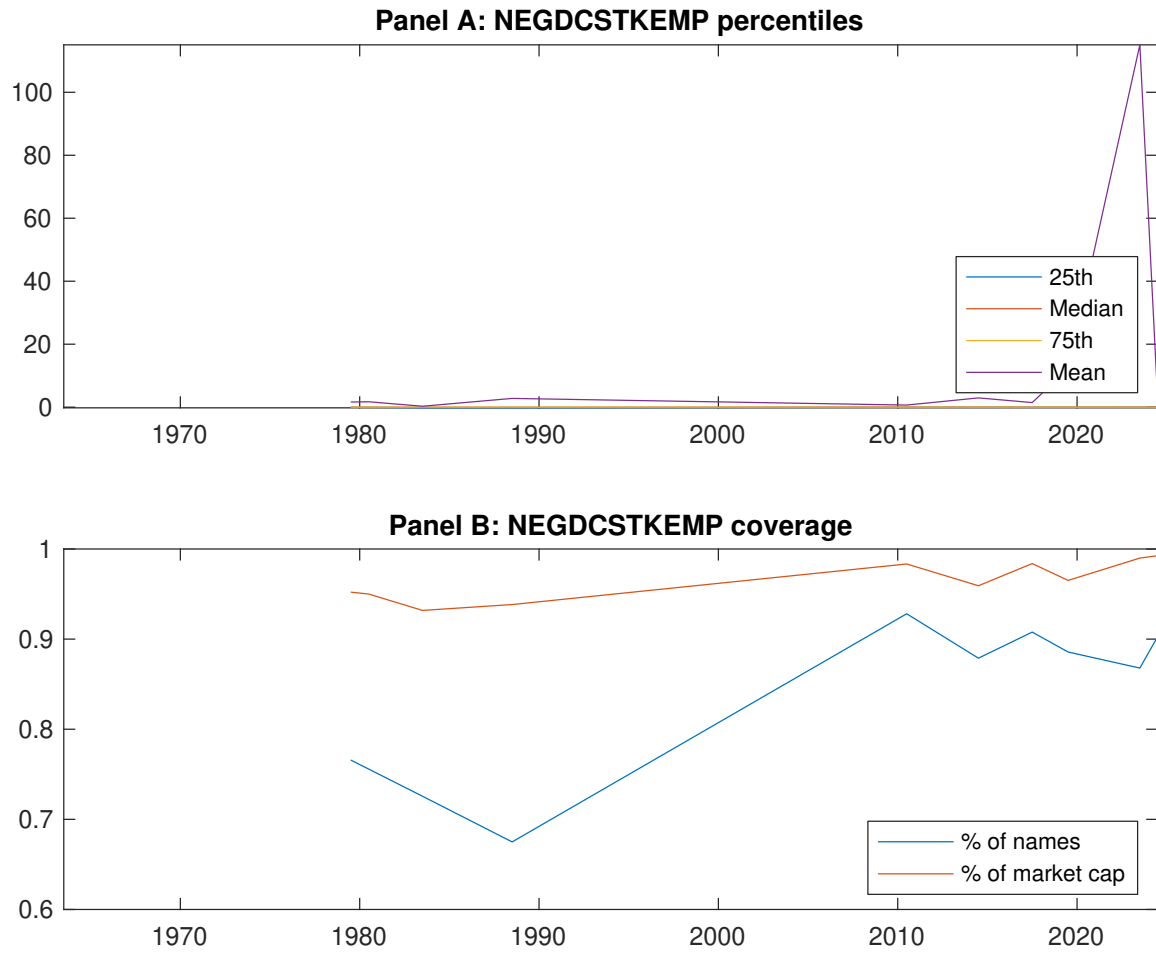


Figure 1: Times series of EDC percentiles and coverage. This figure plots descriptive statistics for EDC. Panel A shows cross-sectional percentiles of EDC over the sample. Panel B plots the monthly coverage of EDC relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EDC. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EDC-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.42 [2.54]	0.58 [3.19]	0.65 [3.57]	0.65 [3.97]	0.76 [4.76]	0.34 [4.45]
α_{CAPM}	-0.14 [-2.72]	-0.05 [-1.11]	0.03 [0.54]	0.10 [1.90]	0.22 [4.56]	0.36 [4.64]
α_{FF3}	-0.15 [-2.95]	-0.03 [-0.75]	0.05 [0.91]	0.04 [0.91]	0.18 [3.98]	0.33 [4.31]
α_{FF4}	-0.13 [-2.53]	0.00 [0.01]	0.03 [0.67]	0.04 [0.89]	0.16 [3.52]	0.30 [3.77]
α_{FF5}	-0.15 [-2.84]	0.02 [0.56]	0.06 [1.11]	-0.04 [-0.87]	0.09 [2.02]	0.24 [3.08]
α_{FF6}	-0.13 [-2.53]	0.05 [1.09]	0.05 [0.88]	-0.03 [-0.69]	0.08 [1.85]	0.22 [2.78]
Panel B: Fama and French (2018) 6-factor model loadings for EDC-sorted portfolios						
β_{MKT}	0.96 [76.58]	1.04 [97.27]	1.03 [81.44]	1.00 [89.93]	0.98 [89.79]	0.02 [0.93]
β_{SMB}	-0.08 [-4.17]	0.02 [1.51]	0.06 [3.43]	-0.04 [-2.63]	-0.02 [-1.41]	0.05 [1.98]
β_{HML}	0.09 [3.62]	-0.02 [-1.19]	-0.04 [-1.56]	0.09 [4.23]	-0.00 [-0.02]	-0.09 [-2.44]
β_{RMW}	0.05 [2.06]	-0.10 [-4.90]	-0.00 [-0.11]	0.15 [6.87]	0.13 [6.25]	0.08 [2.26]
β_{CMA}	-0.10 [-2.88]	-0.09 [-3.14]	-0.06 [-1.67]	0.16 [5.04]	0.24 [7.77]	0.34 [6.46]
β_{UMD}	-0.02 [-1.62]	-0.03 [-3.20]	0.02 [1.31]	-0.01 [-0.97]	0.01 [0.86]	0.03 [1.59]
Panel C: Average number of firms (n) and market capitalization (me)						
n	695	671	564	645	700	
me (\$10 ⁶)	1850	1482	1997	2177	2537	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EDC strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.34 [4.45]	0.36 [4.64]	0.33 [4.31]	0.30 [3.77]	0.24 [3.08]	0.22 [2.78]
Quintile	NYSE	EW	0.53 [7.71]	0.58 [8.61]	0.50 [7.85]	0.42 [6.65]	0.34 [5.65]	0.28 [4.78]
Quintile	Name	VW	0.30 [3.85]	0.31 [3.97]	0.30 [3.79]	0.28 [3.49]	0.23 [2.89]	0.22 [2.76]
Quintile	Cap	VW	0.25 [3.27]	0.26 [3.36]	0.25 [3.23]	0.21 [2.70]	0.18 [2.34]	0.16 [2.02]
Decile	NYSE	VW	0.30 [3.20]	0.27 [2.85]	0.24 [2.58]	0.21 [2.17]	0.20 [2.10]	0.17 [1.84]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.30 [3.98]	0.32 [4.18]	0.29 [3.85]	0.27 [3.56]	0.22 [2.92]	0.21 [2.75]
Quintile	NYSE	EW	0.34 [4.62]	0.41 [5.62]	0.33 [4.84]	0.29 [4.28]	0.18 [2.79]	0.15 [2.44]
Quintile	Name	VW	0.26 [3.38]	0.27 [3.53]	0.26 [3.33]	0.25 [3.17]	0.20 [2.65]	0.20 [2.58]
Quintile	Cap	VW	0.21 [2.80]	0.22 [2.92]	0.21 [2.77]	0.19 [2.48]	0.16 [2.12]	0.15 [1.94]
Decile	NYSE	VW	0.26 [2.77]	0.22 [2.40]	0.20 [2.14]	0.18 [1.92]	0.16 [1.78]	0.15 [1.65]

Table 3: Conditional sort on size and EDC

This table presents results for conditional double sorts on size and EDC. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EDC. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EDC and short stocks with low EDC. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EDC Quintiles					EDC Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.45 [1.73]	0.70 [2.64]	0.83 [3.31]	0.93 [3.75]	1.03 [4.51]	0.61 [6.55]	0.67 [7.31]	0.61 [6.90]	0.55 [6.14]	0.42 [4.99]	0.38 [4.50]
	(2)	0.57 [2.50]	0.74 [3.03]	0.87 [3.69]	0.89 [3.91]	0.96 [4.46]	0.40 [4.50]	0.44 [4.96]	0.37 [4.24]	0.32 [3.63]	0.26 [3.02]	0.23 [2.64]
	(3)	0.52 [2.67]	0.62 [2.81]	0.83 [3.78]	0.81 [3.91]	0.93 [4.80]	0.41 [5.04]	0.42 [5.13]	0.38 [4.69]	0.35 [4.19]	0.29 [3.50]	0.27 [3.20]
	(4)	0.50 [2.77]	0.64 [3.15]	0.78 [3.82]	0.76 [4.02]	0.79 [4.38]	0.29 [3.72]	0.29 [3.65]	0.23 [3.00]	0.23 [2.86]	0.06 [0.79]	0.07 [0.92]
	(5)	0.48 [2.96]	0.49 [2.71]	0.52 [2.96]	0.52 [3.15]	0.71 [4.48]	0.23 [2.55]	0.23 [2.49]	0.23 [2.47]	0.20 [2.19]	0.19 [2.07]	0.18 [1.91]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EDC Quintiles					EDC Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	361	360	359	359	353	31	33	37	29	28	
	(2)	105	103	104	104	103	56	55	56	55	54	
	(3)	76	77	75	76	76	97	96	96	100	99	
	(4)	64	64	64	64	64	200	205	209	211	215	
(5)	59	59	59	59	59	1436	1551	1565	1649	1888		

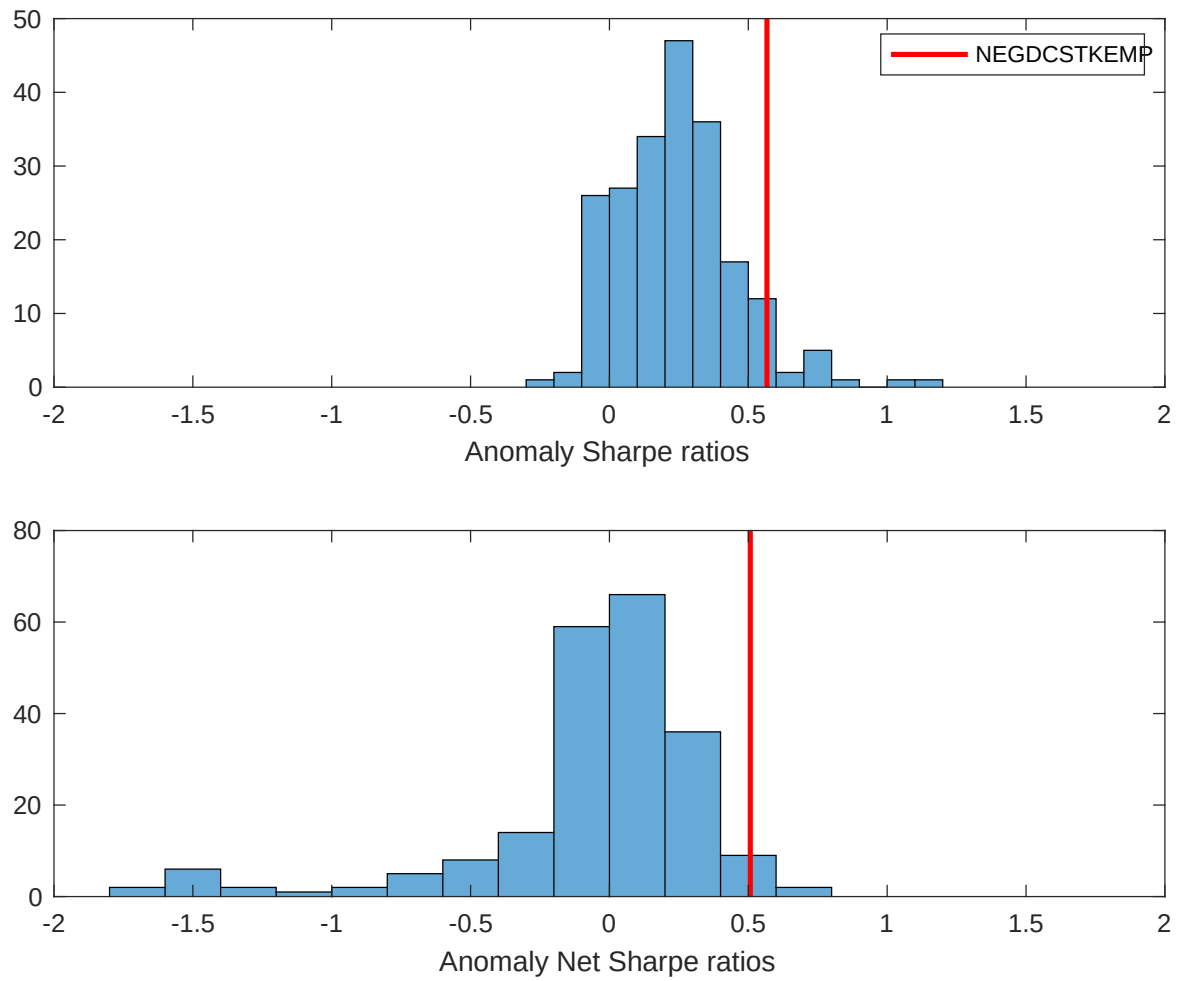


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EDC with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

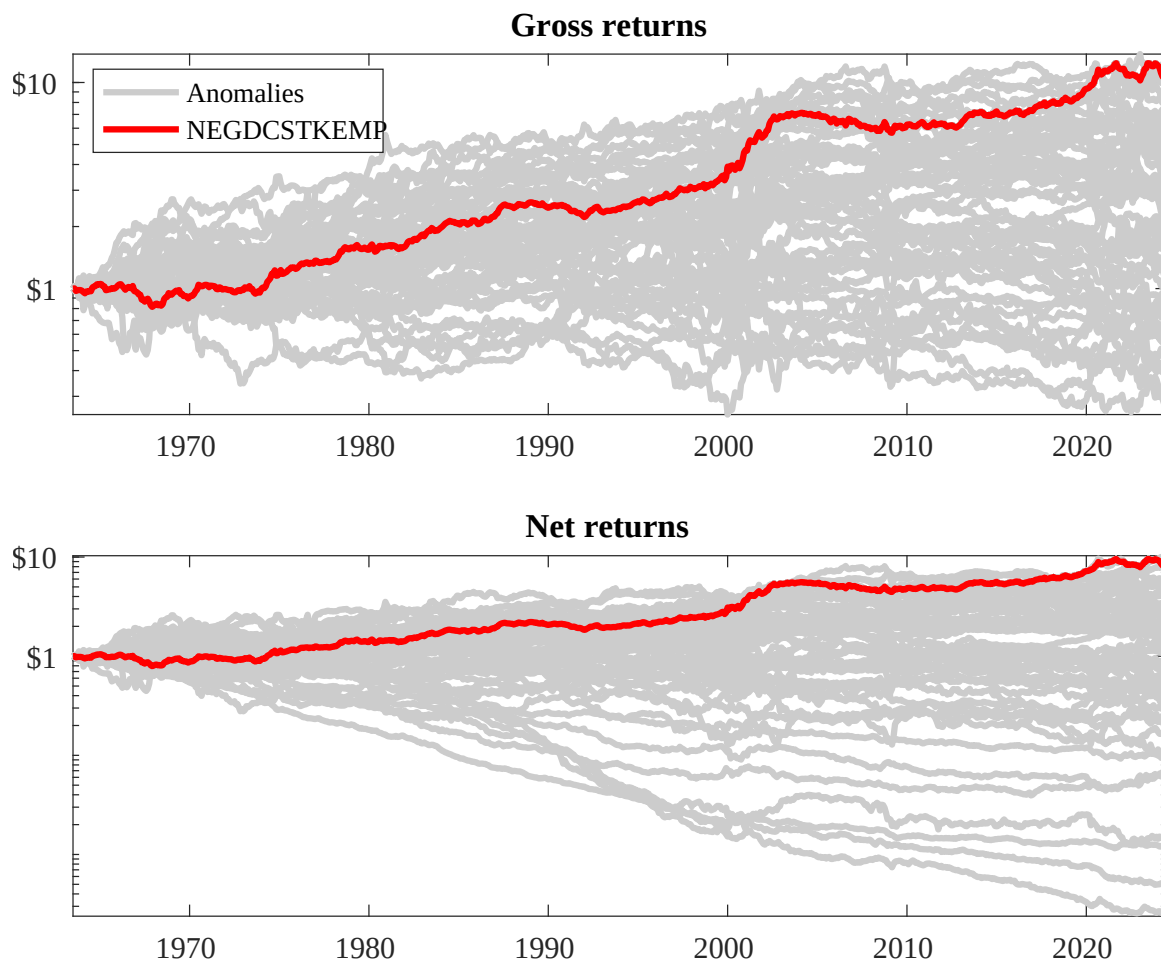
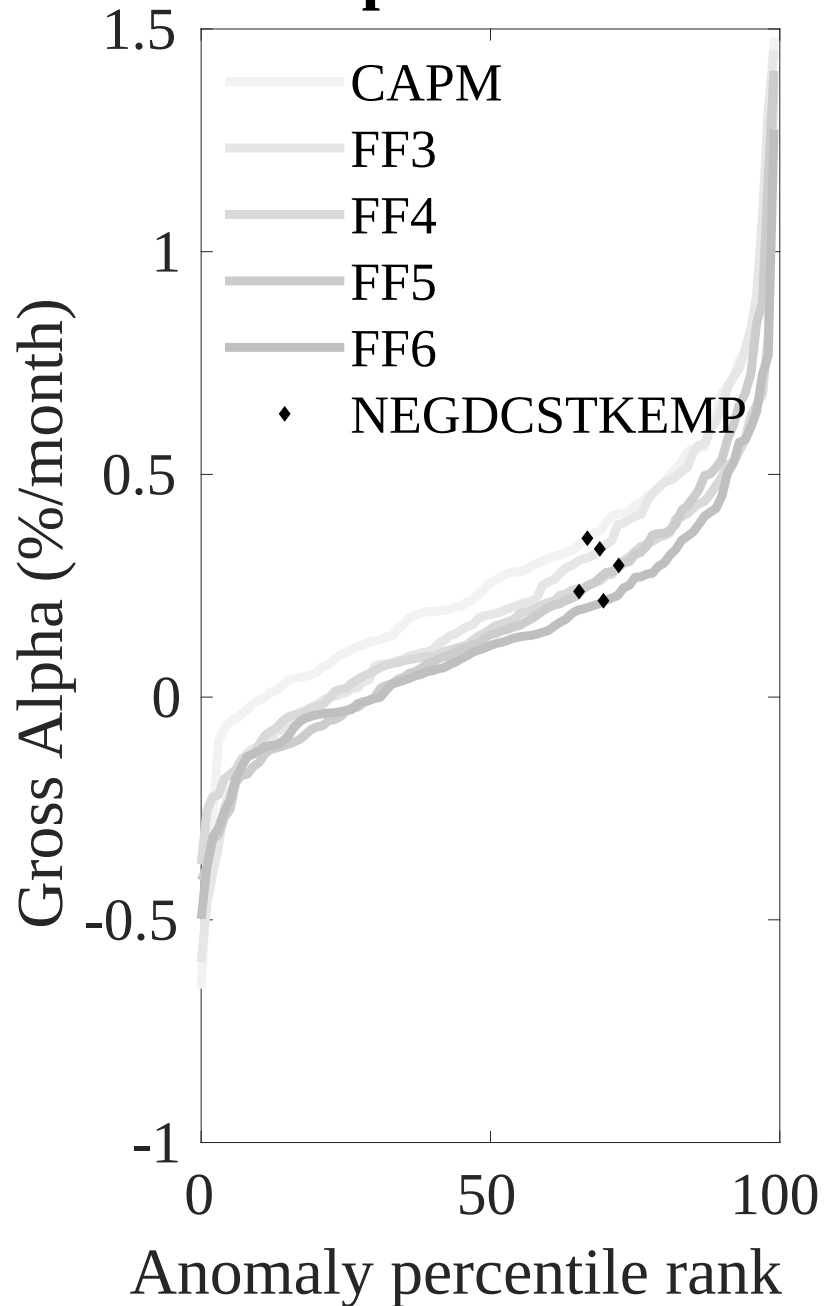


Figure 3: Dollar invested.

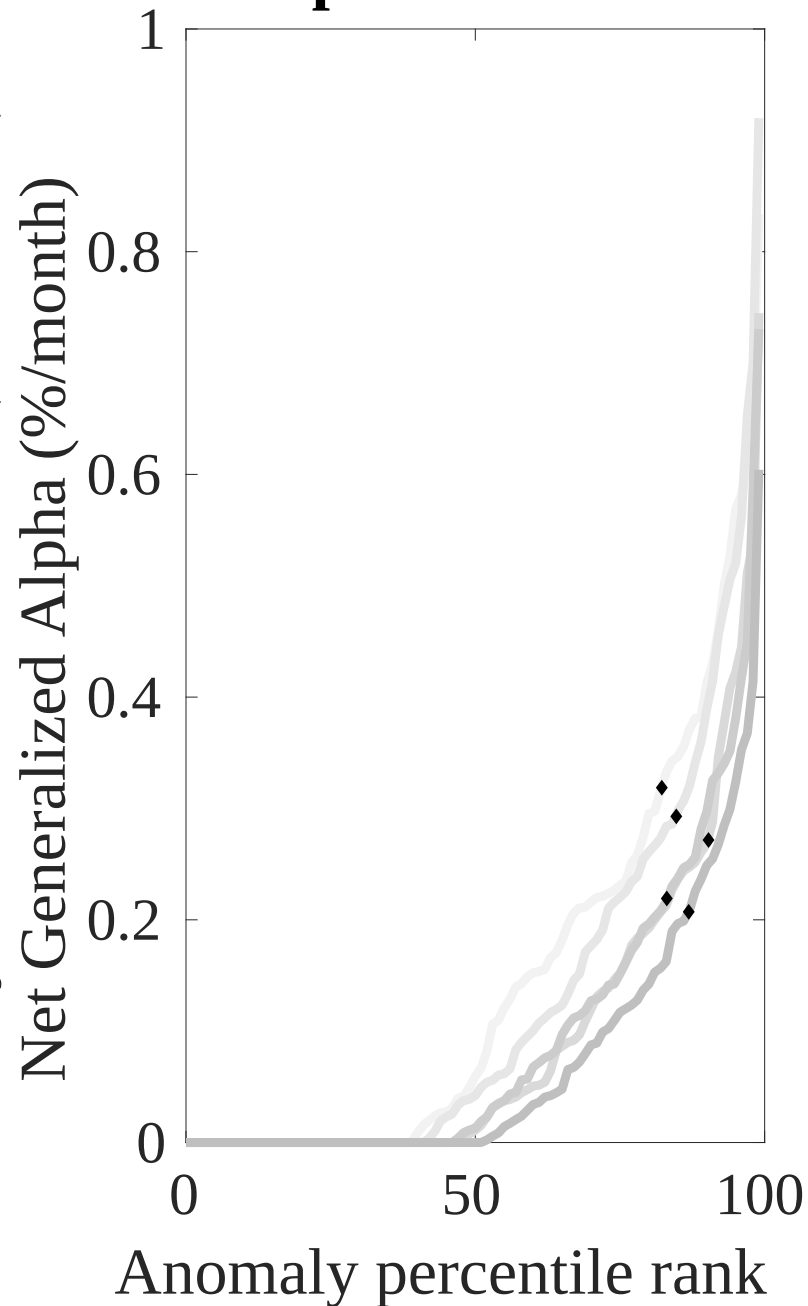
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EDC trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



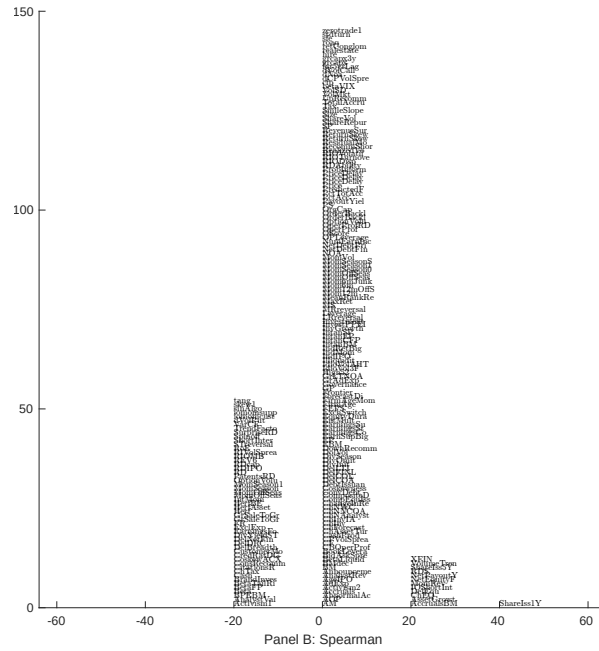
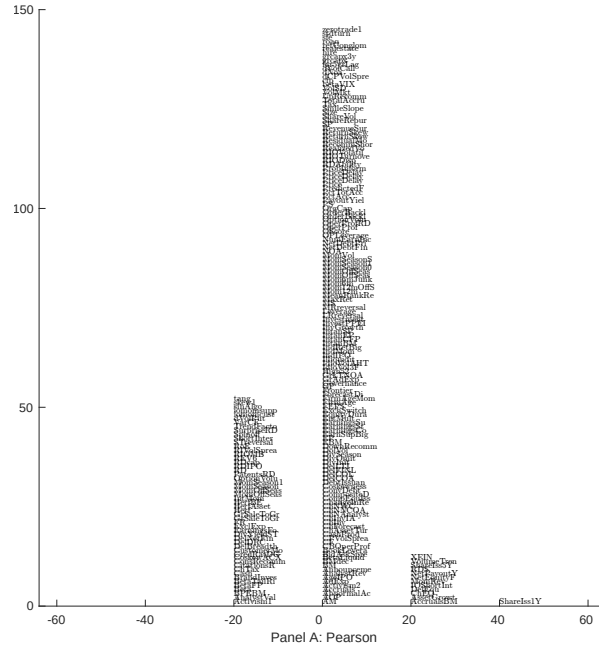


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with EDC. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

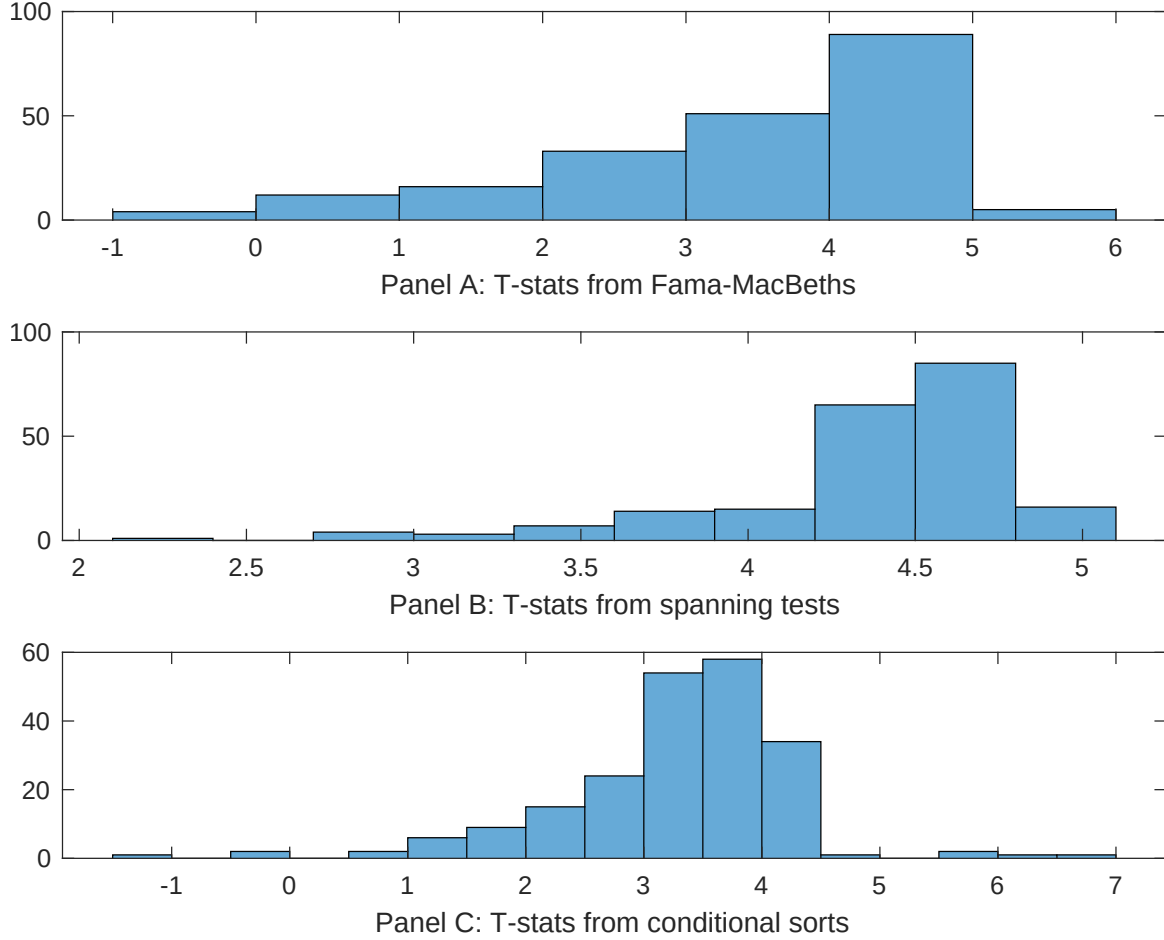


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EDC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDC}EDC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EDC. Stocks are finally grouped into five EDC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EDC. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EDC}EDC_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (1 year), Share issuance (5 year), Growth in book equity, Net Payout Yield, Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [5.99]	0.13 [6.41]	0.17 [6.78]	0.12 [5.42]	0.12 [5.65]	0.13 [6.11]	0.11 [3.67]
EDC	0.47 [3.53]	0.44 [3.12]	0.51 [3.93]	0.46 [3.39]	0.53 [4.10]	0.46 [3.60]	0.31 [2.34]
Anomaly 1	0.14 [5.66]						0.73 [2.53]
Anomaly 2		0.27 [4.89]					0.23 [4.23]
Anomaly 3			0.39 [3.08]				-0.22 [-0.98]
Anomaly 4				0.20 [1.69]			0.98 [0.90]
Anomaly 5					0.12 [3.08]		0.67 [0.11]
Anomaly 6						0.92 [7.49]	0.64 [5.39]
# months	715	715	720	715	720	720	715
$\bar{R}^2(\%)$	0	1	0	1	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EDC trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EDC} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (1 year), Share issuance (5 year), Growth in book equity, Net Payout Yield, Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.20 [2.63]	0.15 [1.93]	0.20 [2.67]	0.24 [3.17]	0.21 [2.78]	0.21 [2.71]	0.15 [2.12]
Anomaly 1	27.66 [7.29]						13.31 [2.91]
Anomaly 2		27.18 [6.75]					12.79 [2.82]
Anomaly 3			32.00 [7.95]				38.42 [6.60]
Anomaly 4				15.70 [5.32]			4.21 [1.32]
Anomaly 5					12.84 [3.36]		-11.62 [-2.17]
Anomaly 6						-0.25 [-0.05]	-16.29 [-3.25]
mkt	2.57 [1.44]	3.12 [1.73]	2.95 [1.66]	4.33 [2.32]	1.66 [0.91]	1.90 [1.03]	4.87 [2.73]
smb	7.70 [2.98]	8.42 [3.23]	5.54 [2.16]	9.14 [3.42]	6.45 [2.43]	6.64 [2.46]	8.94 [3.48]
hml	-8.11 [-2.35]	-5.19 [-1.49]	-11.32 [-3.27]	-13.45 [-3.63]	-9.07 [-2.53]	-7.32 [-2.05]	-11.11 [-3.14]
rmw	-6.65 [-1.64]	-1.41 [-0.37]	9.81 [2.83]	-0.81 [-0.21]	9.47 [2.64]	8.30 [2.30]	-4.98 [-1.18]
cma	16.12 [2.90]	20.11 [3.70]	2.33 [0.37]	20.05 [3.52]	20.40 [3.15]	33.63 [4.40]	9.59 [1.25]
umd	1.84 [1.04]	1.33 [0.74]	2.54 [1.45]	4.36 [2.41]	3.43 [1.88]	3.14 [1.71]	0.53 [0.30]
# months	728	728	732	728	732	732	728
$\bar{R}^2(\%)$	13	12	14	10	8	7	21

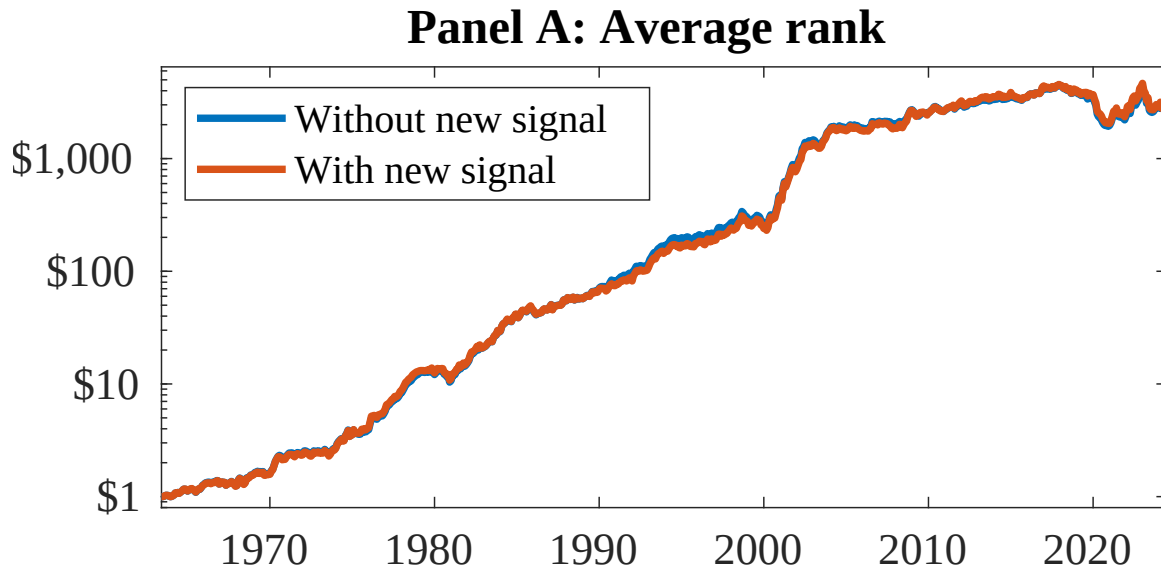


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EDC. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

References

- Bansal, R. and Yaron, A. (2004). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.
- Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.
- Campbell, J. Y. and Cochrane, J. H. (1999). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Daniel, K. and Titman, S. (2006). Market reactions to tangible and intangible information. *Journal of Finance*, 61(4):1605–1643.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Eisfeldt, A. L. and Papanikolaou, D. (2013). Organization capital and the cross-section of expected returns. *Journal of Finance*, 68(4):1365–1406.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Gabaix, X. (2012). Variable rare disasters: An exactly solved framework for ten puzzles in macro-finance. *Quarterly Journal of Economics*, 127(2):645–700.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.
- Pontiff, J. and Woodgate, A. (2008). Share issuance and cross-sectional returns. *Journal of Finance*, 63(2):921–945.